

1. Administrative & Study Identification

Full Study Title: A Study to Evaluate the Effect of R07204239 on Insulin Sensitivity and Muscle Composition in Participants With Type 2 Diabetes Mellitus (T2DM) and Overweight or Obesity **Short Title/Acronym:** R07204239 T2DM/Obesity Study **Protocol Number:** BP45980 **NCT Number:** NCT07137585 **Sponsor Name:** F. Hoffmann-La Roche AG **Investigational Product:** R07204239 (Emugrobart) **Phase of Development:** Phase I **Medical Monitor:** Roche Clinical Trial Contact (global-roche-genentech-trials@gene.com / 888-662-6728)

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the effect of R07204239 on insulin sensitivity versus placebo in participants with Type 2 Diabetes Mellitus (T2DM) and obesity or overweight. **Secondary Objectives:** To evaluate muscle composition and function, pharmacokinetic profile, and overall safety and tolerability of R07204239.

2.2 Background & Rationale Diabetes mellitus is a group of health conditions that cause a person's blood sugar to become too high. This happens when the body does not make enough insulin or does not respond to insulin the way it should. Treatment with R07204239 alongside diet modification and exercise might lead to improved insulin sensitivity and weight loss. This study evaluates whether R07204239 can provide clinical benefits for this population over a 24-week period.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Quadruple Blind (Double-blind) **Randomization:** Randomized (Parallel Assignment)

3.2 Planned Enrollment & Duration Target N: 30 participants **Number of Sites:** 1 location (Neuss, North Rhine-Westphalia, Germany) **Estimated Study Start:** 30-Sep-2025 **Estimated Primary Completion:** 13-Sep-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Confirmed diagnosis of Type 2 Diabetes Mellitus (T2DM). 3. Overweight or obese classification.

4.2 Exclusion Criteria 1. Type 1 diabetes mellitus, known latent autoimmune diabetes in adults, or episode of ketoacidosis/hyperosmolar state requiring hospitalisation within 6 months prior to screening. 2. Active proliferative diabetic retinopathy, diabetic maculopathy, or severe non-proliferative diabetic retinopathy requiring acute treatment. 3. Severe hypoglycaemia within 6 months prior to the screening visit.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: R07204239 or matching placebo, administered every 4 weeks (Q4W). **Route of Administration:** Subcutaneous (SC) injection (in the belly). **Duration of Treatment:** 24 weeks.

5.2 Concomitant & Prohibited Therapy Permitted: Standard diet modification and exercise recommendations. **Prohibited:** Central nervous system stimulants (e.g., Ritalin-SR), with the exception of caffeinated beverages.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Baseline Assessments
Baseline	Day 0	Randomization, First SC Dose of R07204239 or Placebo
Interim	Weeks 4, 8, 12, 16, 20	Safety Labs, Efficacy Assessment, Q4W SC Dosing
End of Study	Week 24	Final Assessment of Insulin Sensitivity and Muscle Composition, IP Accountability

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Change from baseline in insulin sensitivity versus placebo after 24 weeks of treatment.

Measurement Method: Clinical assessment of blood sugar control and insulin response.

7.2 Secondary & Exploratory Endpoints * Change from baseline in muscle composition and function. * Pharmacokinetic parameters of R07204239. * Frequency and severity of adverse events.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Regular check-ups for known unwanted effects, including local irritation, pain/swelling at the injection site, allergic reactions, and muscle twitching/spasms. **Serious Adverse Events (SAEs):** Routine reporting guidelines as designated by the study protocol. **Data Monitoring**

Committee (DMC): Internal sponsor review committees tracking ongoing safety signals.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy; Safety Set for all participants receiving at least one dose of study medication. **Sample Size Justification:** Target enrollment of $N = 30$ assigned across the active and placebo groups to assess safety and preliminary signals of efficacy. **Handling of Missing Data:** Defined per sponsor's SAP, typically utilizing Mixed-Effect Model Repeated Measure (MMRM) or similar methodologies for continuous variables.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine clinical site monitoring visits at the designated study center. **Audit Trail:** Electronic Case Report Form (eCRF) tracking with source data verification.

11. Study Identifiers & Governance

Primary ID: BP45980 **Registry Number:** NCT07137585 **Sponsor/Lead Organization:** F. Hoffmann-La Roche AG **Study Title:** A Study to Evaluate the Effect of R07204239 on Insulin Sensitivity and Muscle Composition in Participants With Type 2 Diabetes Mellitus (T2DM) and Overweight or Obesity **Official Regulatory Title:** Effects of R07204239 on Insulin Sensitivity, Muscle Composition and Function in Participants Living With Type 2 Diabetes Mellitus and Overweight or Obesity: A Double-blind, Randomized, Placebo-controlled Study

12. Clinical Framework (Metabolic Specific)

Primary Condition: Type 2 Diabetes Mellitus (T2DM), Overweight, Obesity **Diagnosis Threshold:** Clinically confirmed T2DM with overweight or obese BMI **Medical Methodology:** Investigational monoclonal antibody therapy combined with lifestyle/diet modification **Optimization Target:** Improved insulin sensitivity and weight loss

13. Study Procedures & Methodology

Management Pathway: Standard metabolic tracking coupled with Q4W SC injections **Education Protocol (HCP):** Site initiation and

ongoing trial management **Education Protocol (Patient):** Guidance on diet modification, exercise, and reporting of expected injection-site reactions **Quality Audit Mechanism:** Investigator reporting and sponsor oversight

14. Observation & Follow-Up Schedule

Total Duration: 24 weeks of core treatment **Onsite Visit Cadence:** Every 4 weeks (Q4W) for injections and monitoring **Remote Monitoring Frequency:** Continuous safety tracking as needed **Checklist Review Interval:** Regular study checkpoints up to Week 24

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				